

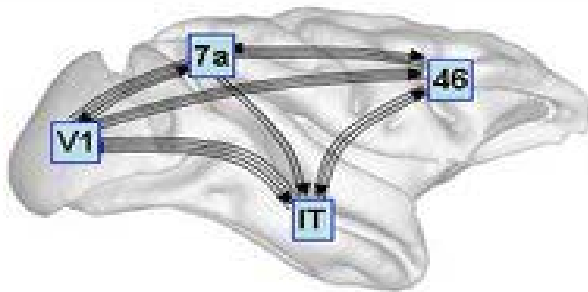
Functional/Effective Connectivity for fMRI

Rik Henson

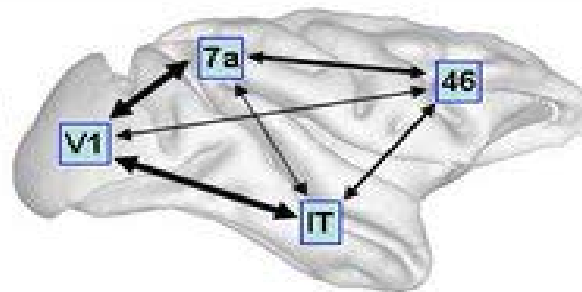
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Structural, functional & effective connectivity

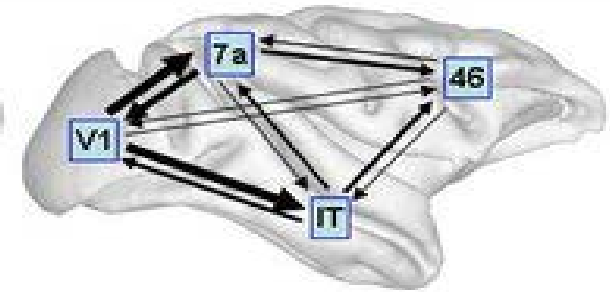
structural connectivity



functional connectivity



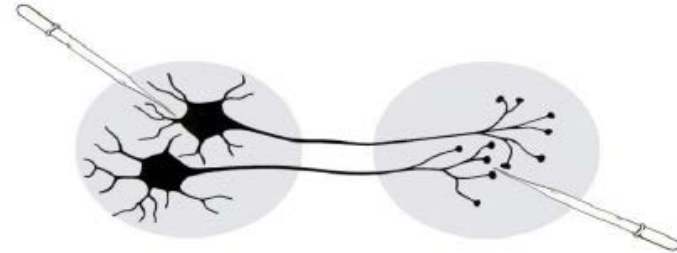
effective connectivity



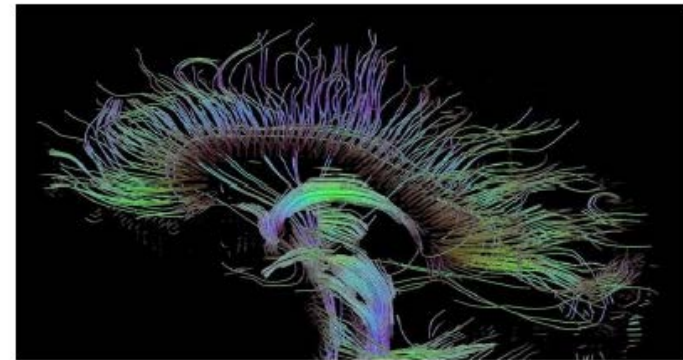
- **Structural/anatomical connectivity**
= presence of axonal connections / white matter tracks (eg, DWI)
- **Functional connectivity**
= statistical dependencies between regional time series (eg, ICA)
- **Effective connectivity**
= causal (direct/directed) interactions between populations (eg, DCM)
(based on explicit network models)

Structural vs Functional connectivity

- Tracing studies

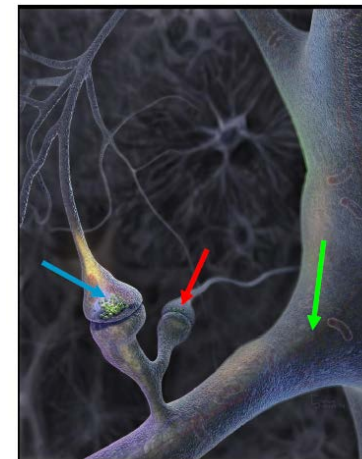


- Tractography from DWI



But functionally, effect of one neuron on another can depend on:

- Activity of a third (gating)
- Rapid changes in plasticity



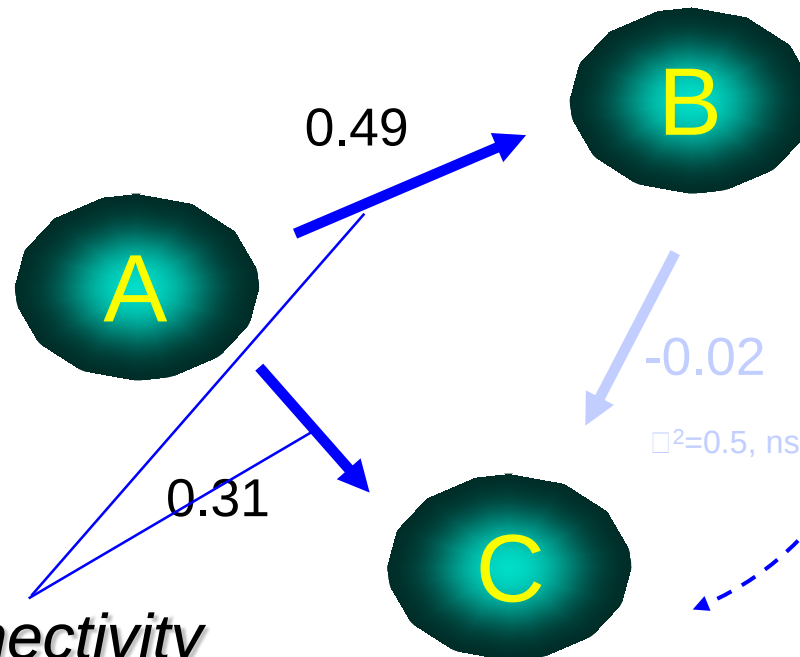
Functional vs Effective connectivity

No connection between B and C, yet B and C correlated because of common input from A, eg:

A = V1 fMRI time-series

$B = 0.5 * A + e1$

$C = 0.3 * A + e2$



Effective connectivity

Correlations:

A	B	C
1		
0.49	1	
0.30	0.12	1

Functional connectivity

Partial Correlation is approximate way

Functional Connectivity (in fMRI)



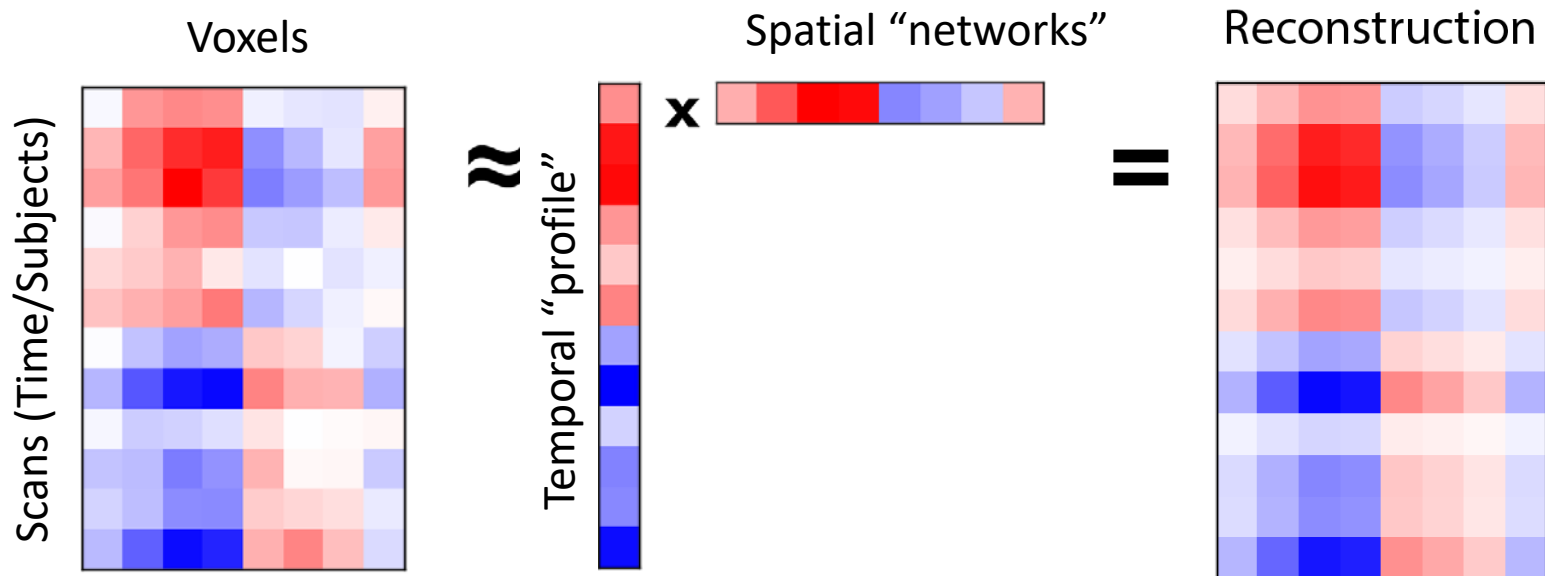
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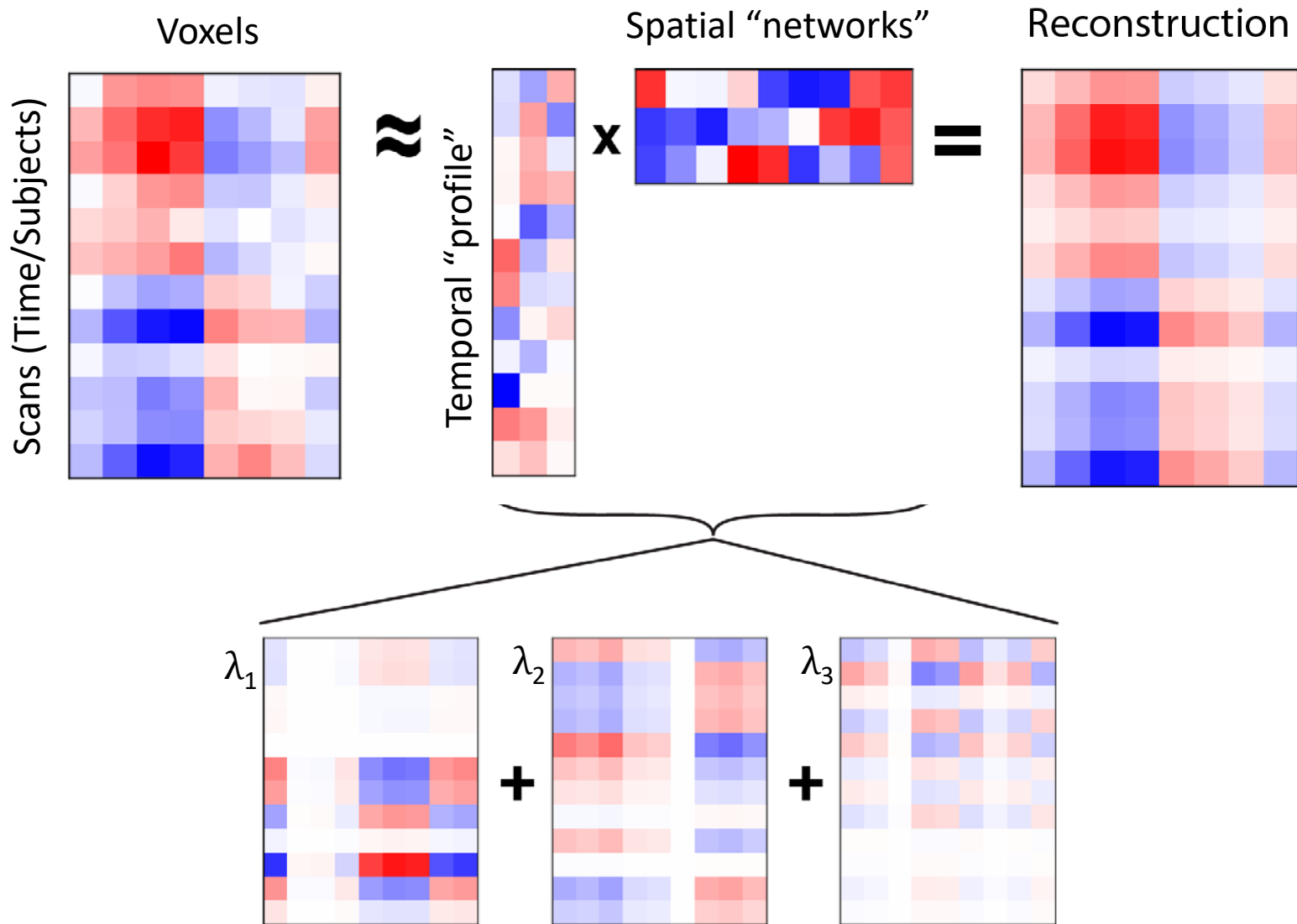
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- State-based connectivity
 - Time-Series Correlation (TSC), eg resting-state/movie, though could be spurious, e.g, cardio/respiratory; motion confounds..., or just reflection of structural connectivity...)
 - Multivariate (whole-brain, voxel-wise) decomposition, eg PCA, ICA
 - ROI(parcel)-based, pairwise connectivity
- Task-Based connectivity
 - Beta-Series Regression (BSR) or Psycho-Physiological Interactions (PPI)
- Network-based models
 - Dynamic Causal Modelling (DCM) (effective connectivity; not covered in notebooks)
 - Graph-theory summaries of functional networks

Matrix Factorisation (eg, SVD)



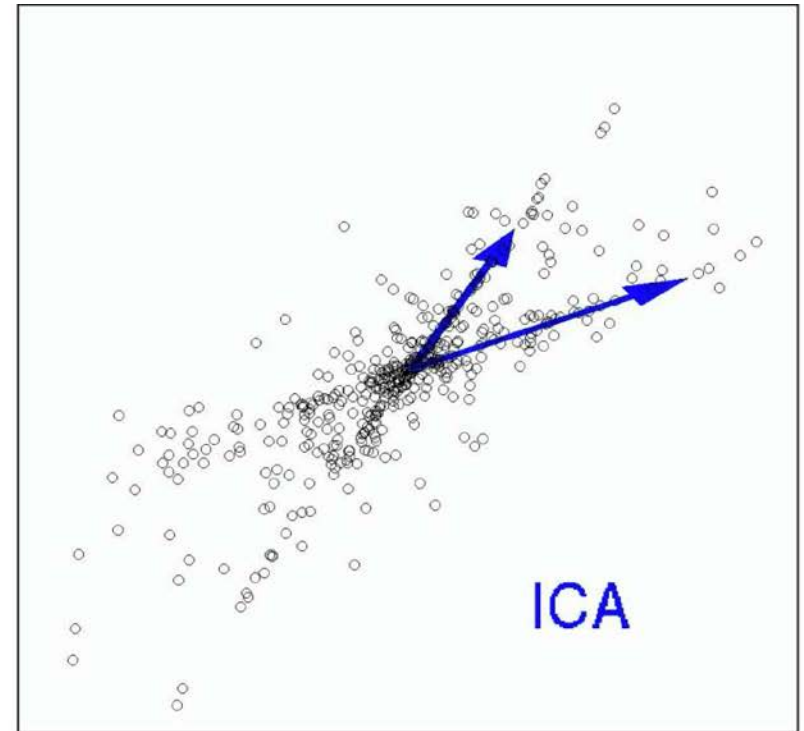
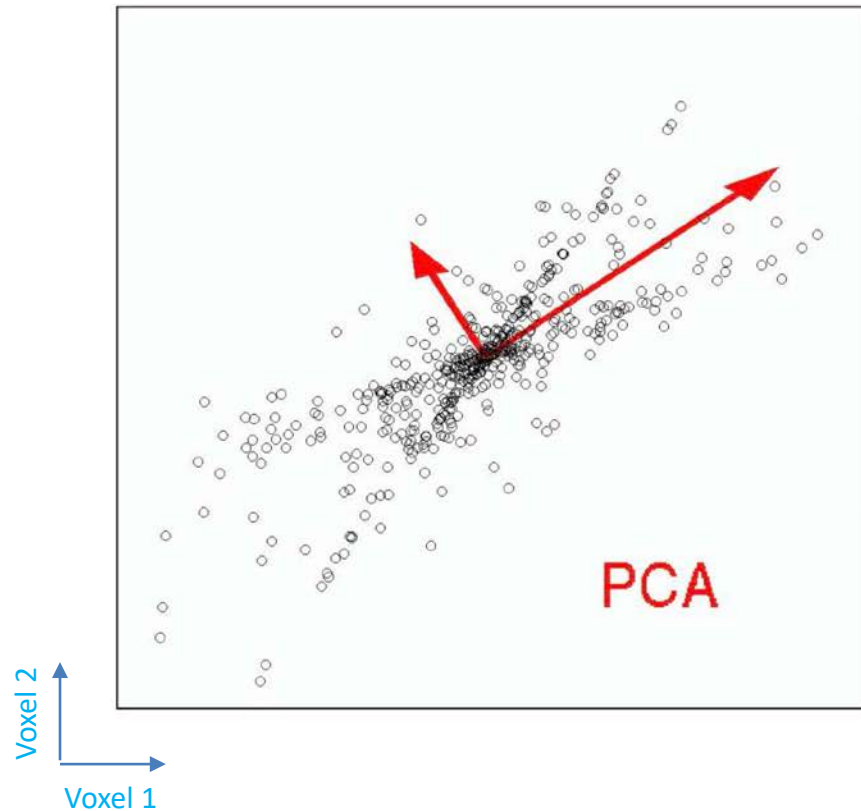
Matrix Factorisation (eg, SVD)



Multivariate fC

- PCA is an SVD of a covariance matrix, assuming *orthogonal* components (e.g, spatial patterns across voxels)
- ICA assumes components are *independent* (usually across space with fMRI, but could be across time) and has been shown more effective in (re)producing characteristic resting-state networks (RSNs), and separating signal from noise

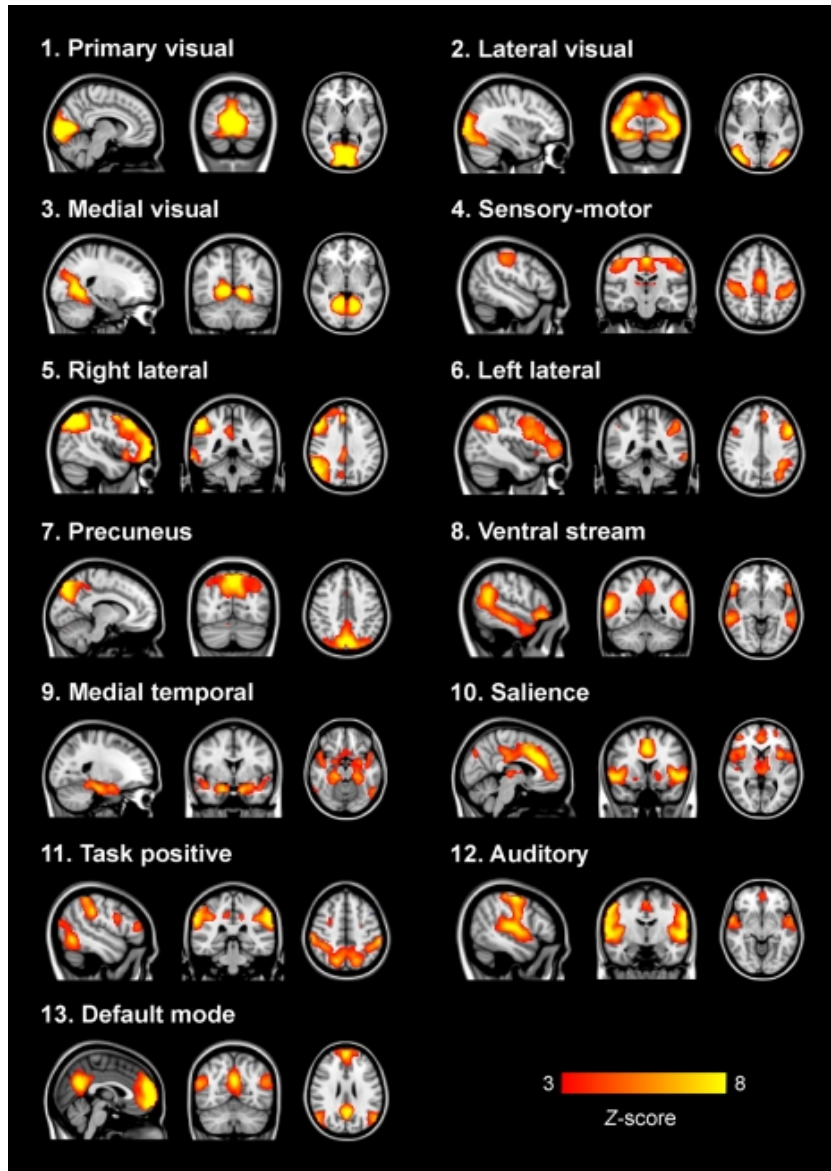
PCA vs ICA



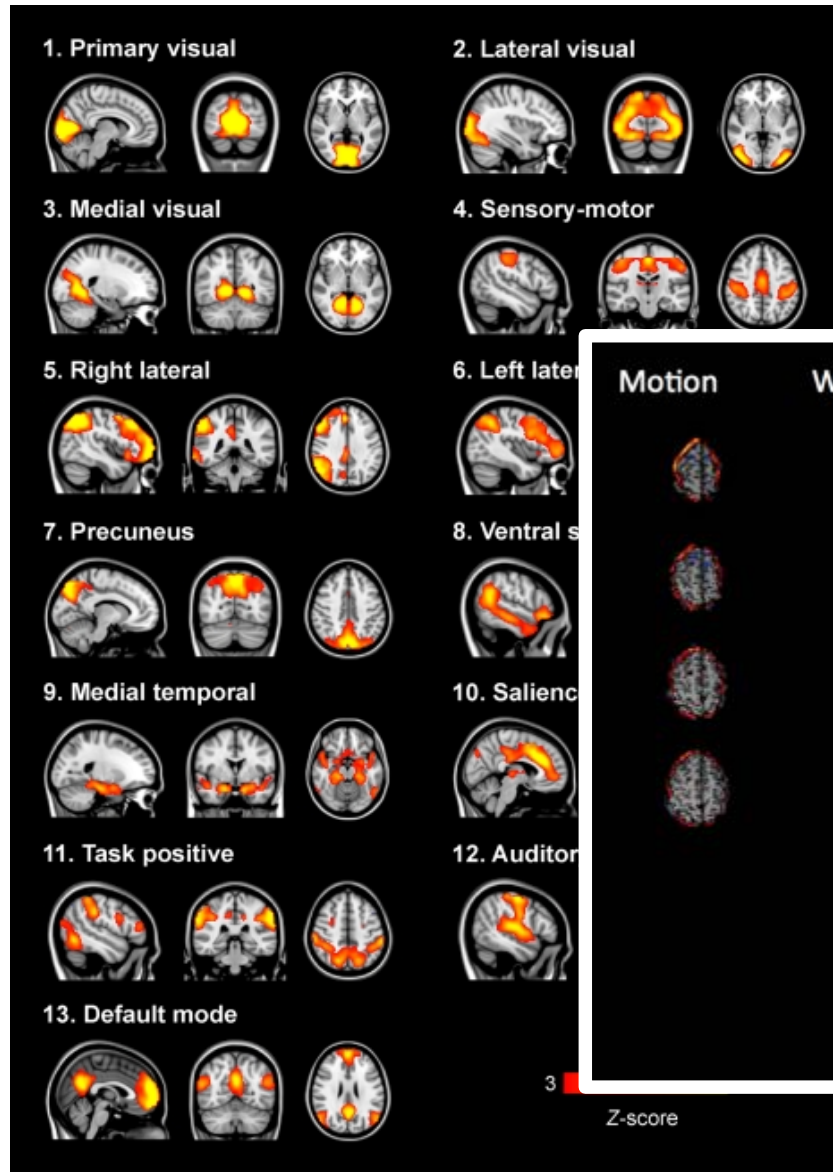
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- There are other types of matrix factorisation (e.g., non-negative, tensor-based) that have been tried, as well as various clustering techniques
- Incidentally, while PCA is an SVD of the data covariance matrix ($Y'Y$), other techniques like CCA and PLS are SVDs of data-model covariance ($Y'X$)...

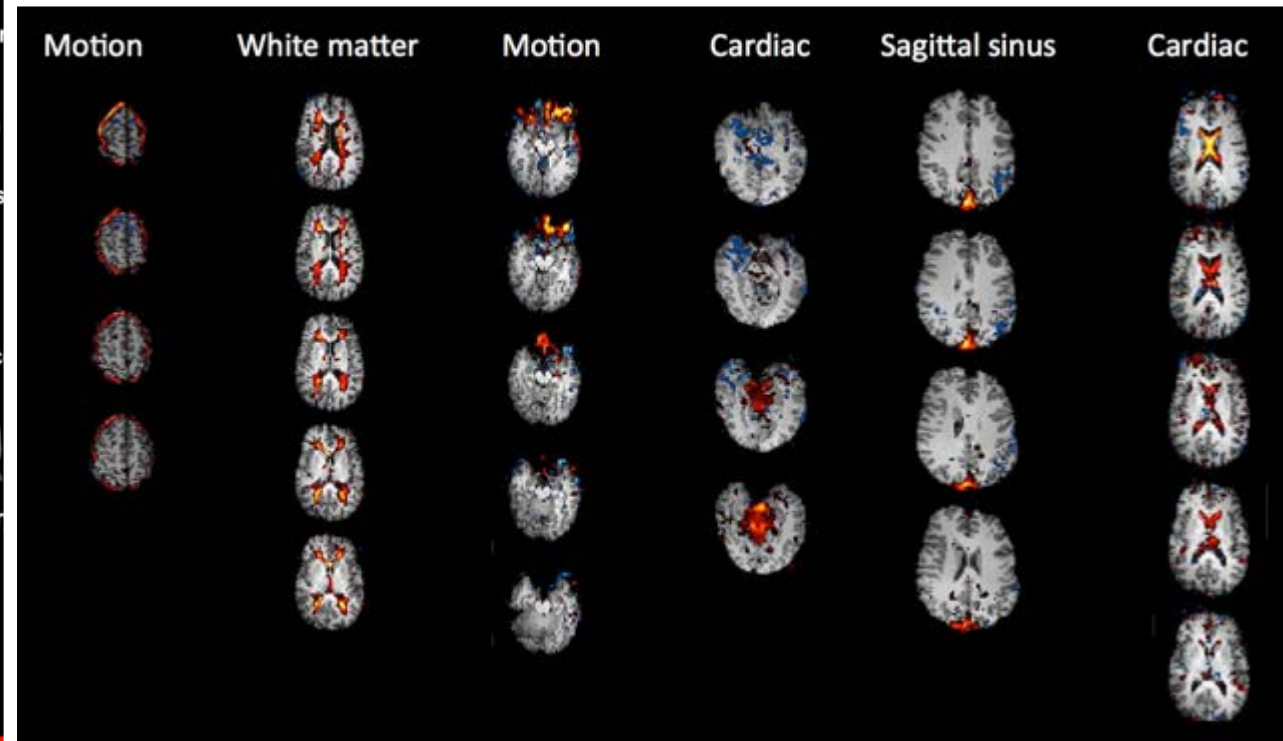
ICA RSNs



ICA RSNs



ICA Noise components



Multivariate fC



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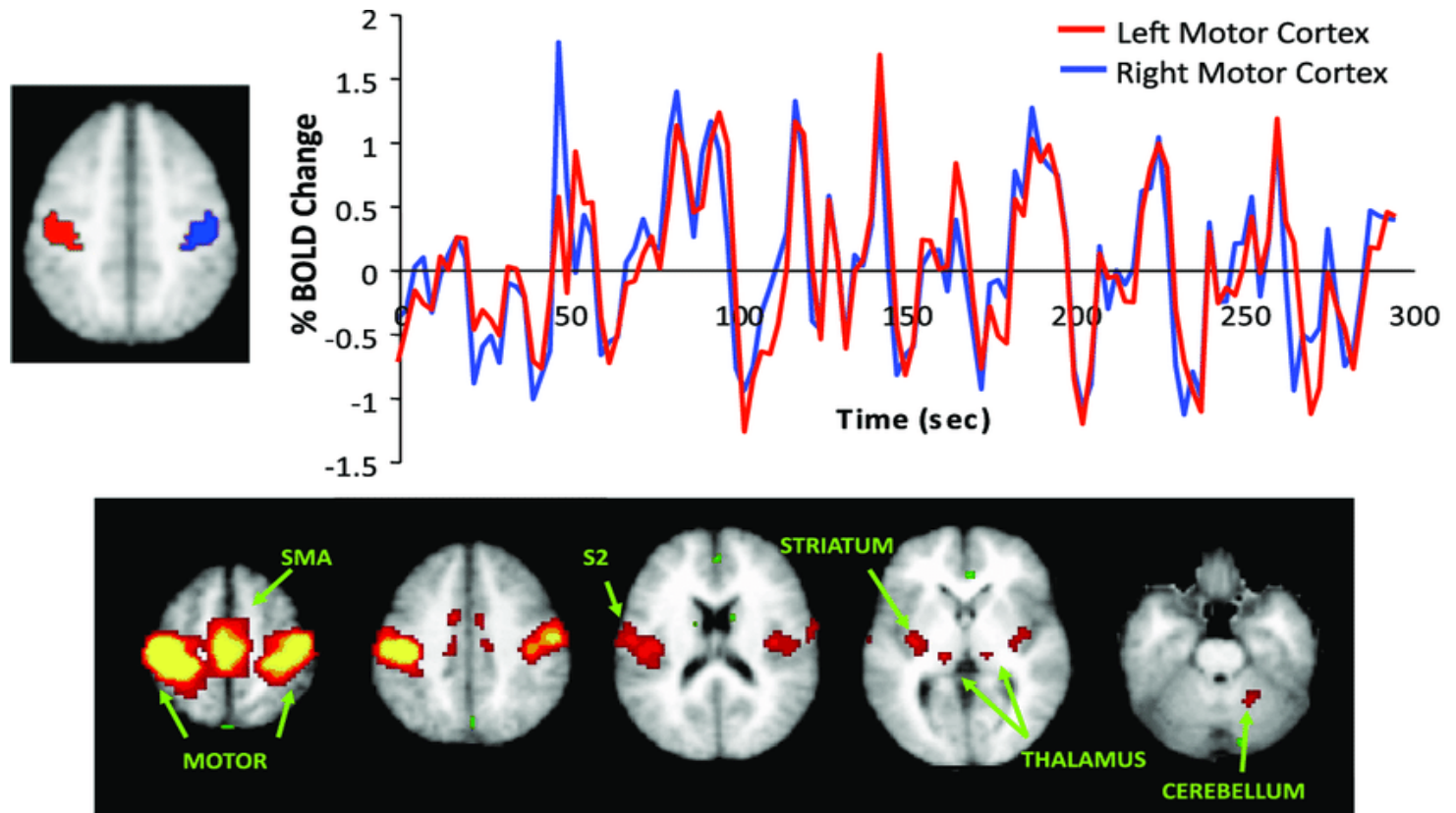


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- Group ICA, concatenate subjects or used Tensor ICA (e.g., , eg MELODIC in FSL: <https://fsl.fmrib.ox.ac.uk/fsl/fslwiki/MELODIC>).
- What signal/noise – eg FSL’ s “FIX”, trained to separate signal and noise (or use multi-echo EPI and remove ICs that not linearly dependent on TE)
- For resting-state, temporal ICs are arbitrary; for subjects performing same task, can correlate the temporal ICs with predicted regressors from design matrix (to get network-level maps)
- For some very stimulating tasks like movie-watching (assumed to activate most of brain), one can correlate the timeseries across subjects for “inter-subject correlation” (ISC), or even “functional normalisation” of brains

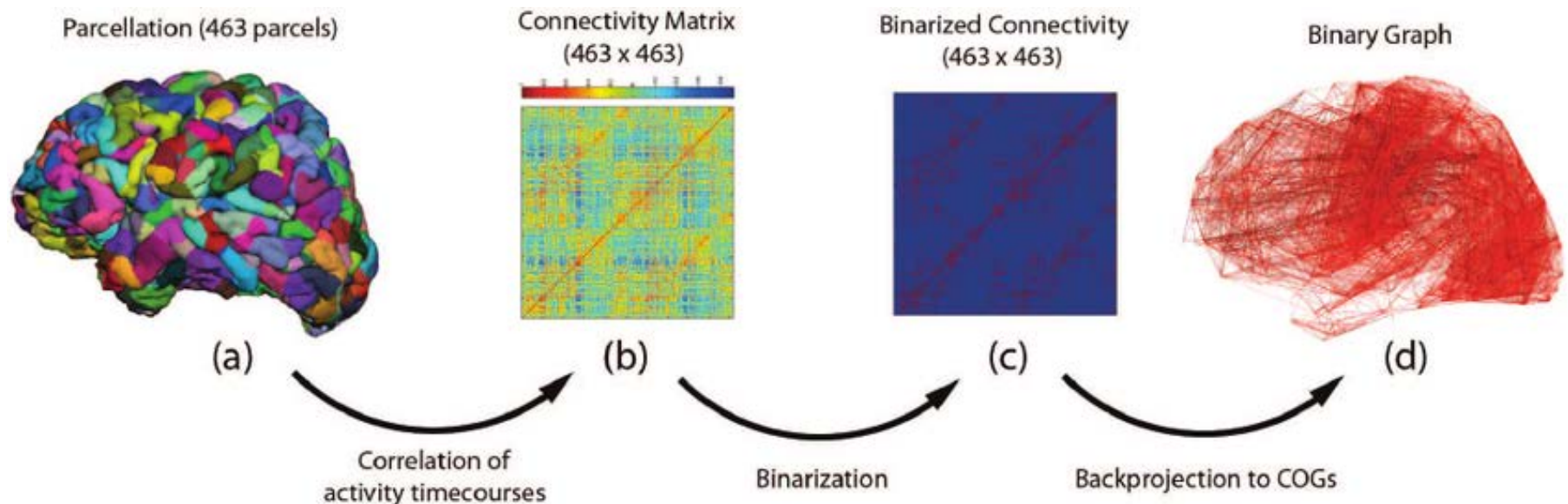
ROI-based fC

- Rather than using a multivariate technique across all voxels, we can calculate pairwise fC between every pair of voxels or ROIs...
- Such as from a “seed” to rest of brain:



ROI-based fC

- Rather than using a multivariate technique across all voxels, we can calculate pairwise fC between every pair of voxels or ROIs...
- Or between every pair of ROIs in an Atlas, to construct a “connectome”:

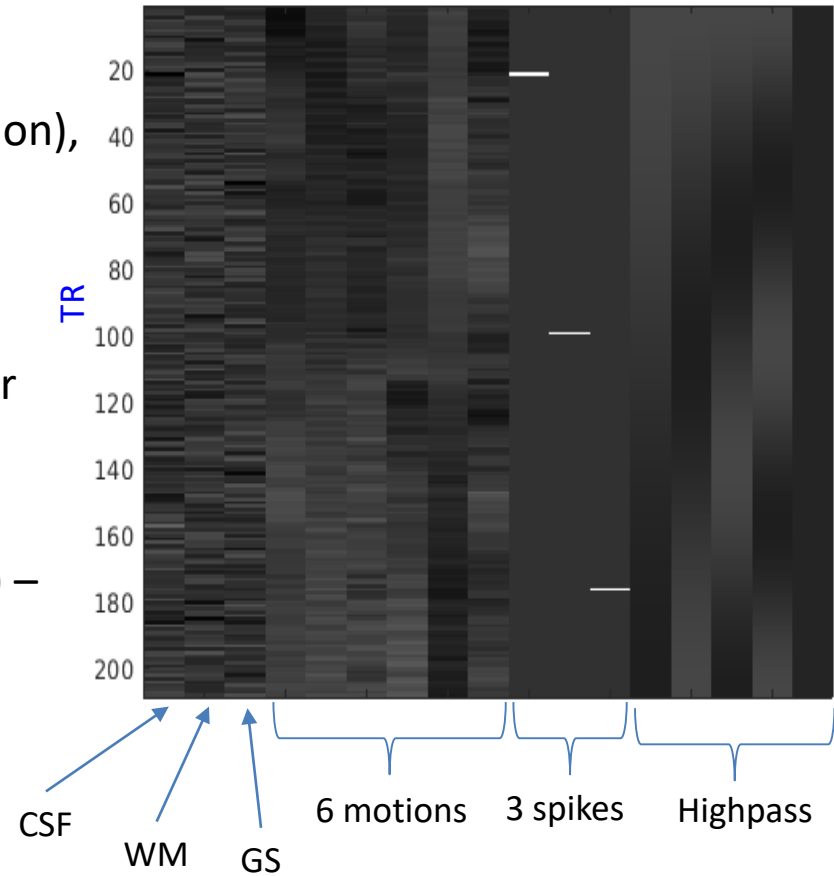


ROI-based fC

- Rather than using a multivariate technique across all voxels, we can calculate pairwise fC between every pair of voxels or ROIs
- A common measure of fC is Pearson's correlation coefficient (though other alternatives include partial correlation, mutual information, Granger, multi-dimensional similarity...)
- However, the timeseries for any pair of voxels/ROIs can have a high correlation owing to non-interesting confounds like motion, breathing, etc...
- ...so we often need to “clean” the data first

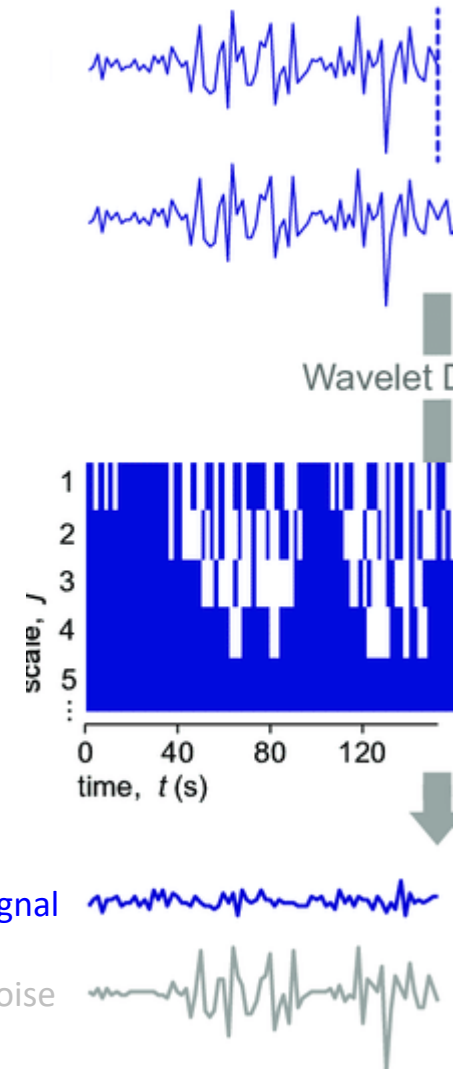
Motion Artefacts

- Add low (band-pass) filter (eg 0.1 Hz)?
- Covary WM/CSF/Global Signal
- “Scrubbing” (removing volumes with high motion), but ignores temporal autocorrelation
- Rather than remove data, regress out motion parameters, including derivatives, second-order expansions (Volterra expansion), etc
- Spikes = extreme values (high order expansion) – model as separate regressors



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- Or define spikes in wavelet domain (but what threshold?)



- High-pass filter, eg to 0.01Hz (since many biorhythms aliased to lower frequency)
- If recorded physiological signals (eg cardiac, respiratory), then can adjust for (Fourier expansion of) those signals (RETROICOR)
 - PhysIO toolbox <https://www.tnu.ethz.ch/en/software/tapas/documentations/physio-toolbox>
- Or can use mean (or first few principal components) of signal in CSF/WM (though care not correlated with GM signal)
 - CONN toolbox (Matlab/SPM): <https://web.conn-toolbox.org/>
- “Global” artifacts, using “global signal regression” (eg across all ROIs)
 - Pros/Cons discussed in <https://doi.org/10.1016/j.neuroimage.2016.11.052>

Task-based fC

- One way to minimise confounds is to compare fC across two or more tasks/conditions. Two main approaches:
 - I. Beta-Series Correlation/Regression (BSR): correlate Betas (trial-wise parameter estimates from single-trial model) across ROIs, and compare correlation coefficients/regression slopes across conditions
 - II. Psycho-Physiological Interactions (PPIs): construct a model (GLM) of fMRI timeseries with regressors for 1) conditions (“psychological”), 2) timeseries from one ROI (seed) (“physiological”) and 3) the interaction between 1+2 (the key PPI term), and test significance of interaction term

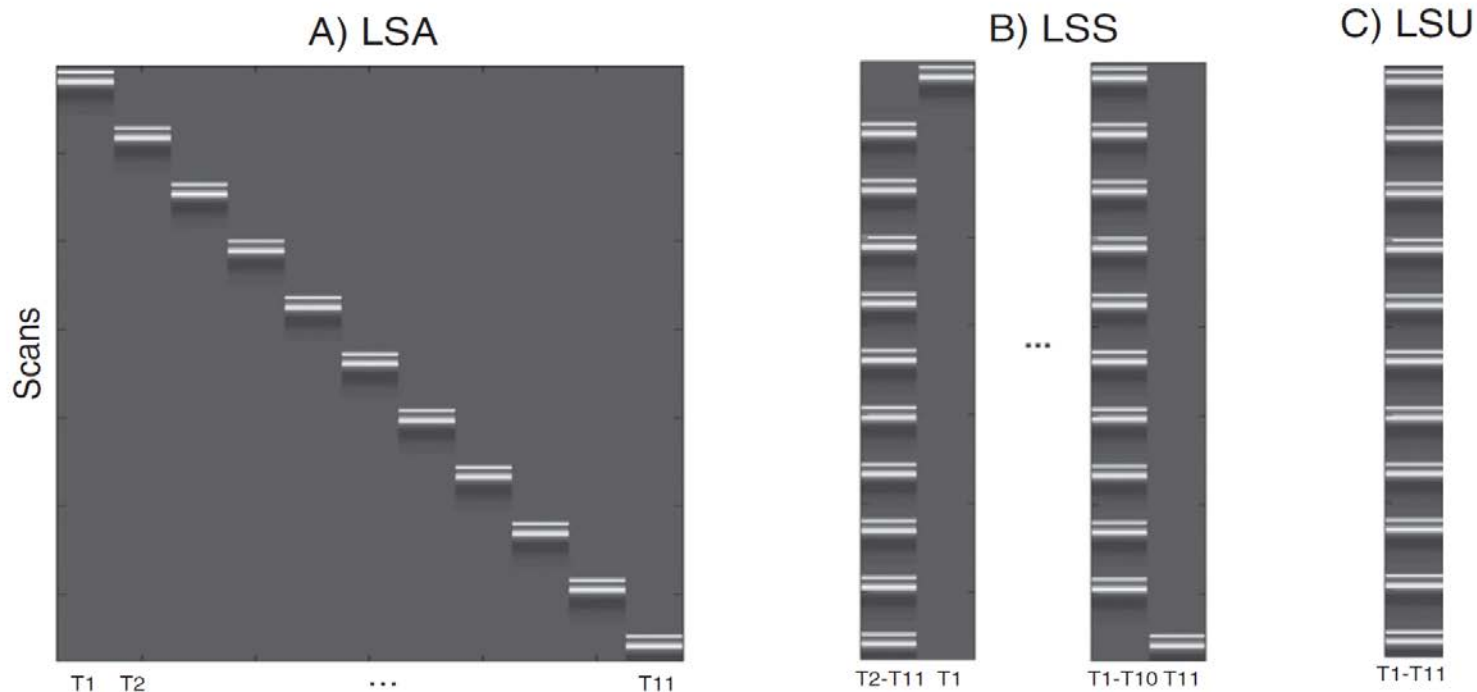
See Masharipov et al 2024 for comparison:

<https://www.nature.com/articles/s42003-024-07088-3>

1. BSR

- One problem with BSR is how to estimate single-trial responses for event-related designs with short SOAs:

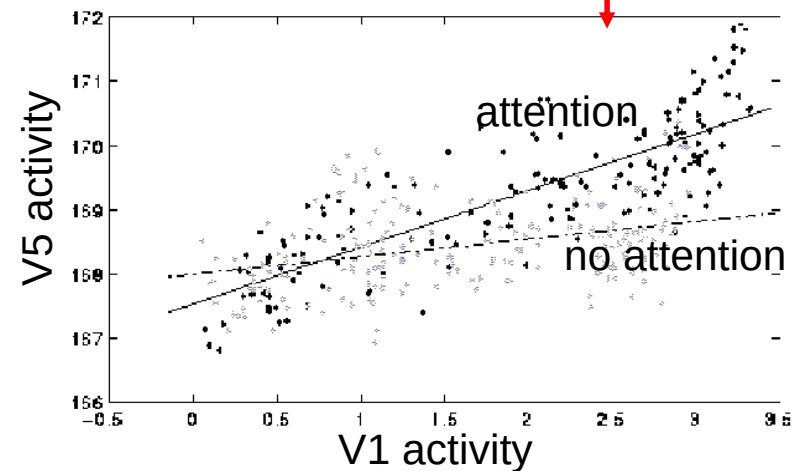
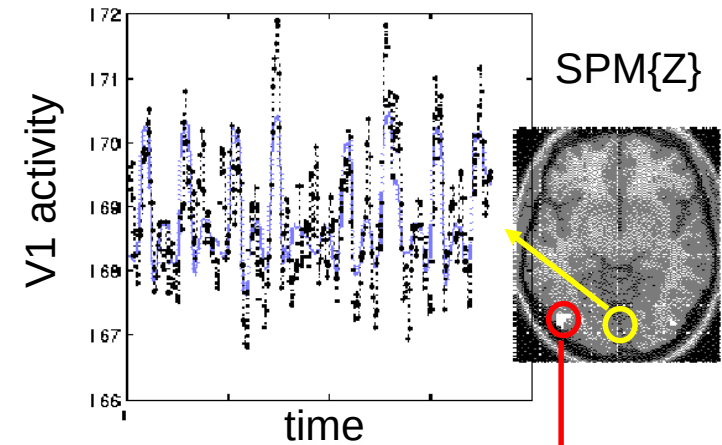
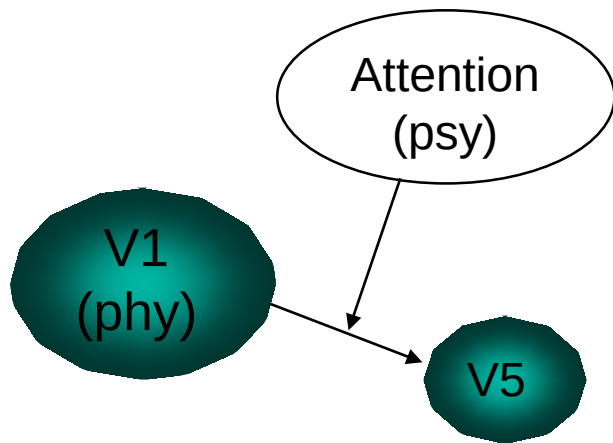
H. Abdulrahman, R.N. Henson / NeuroImage 125 (2016) 756–766



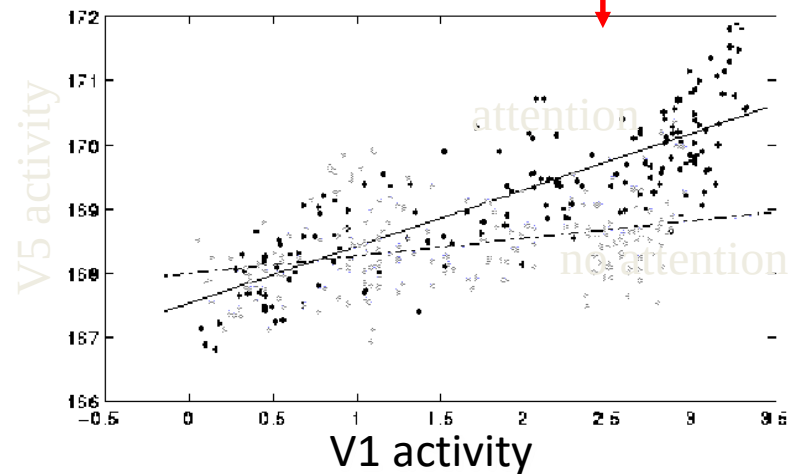
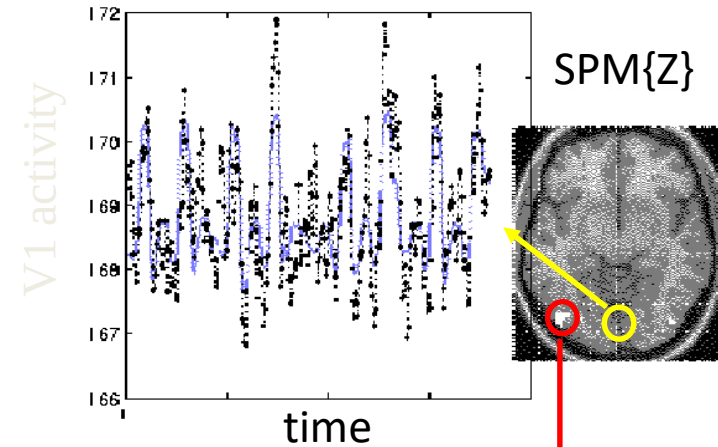
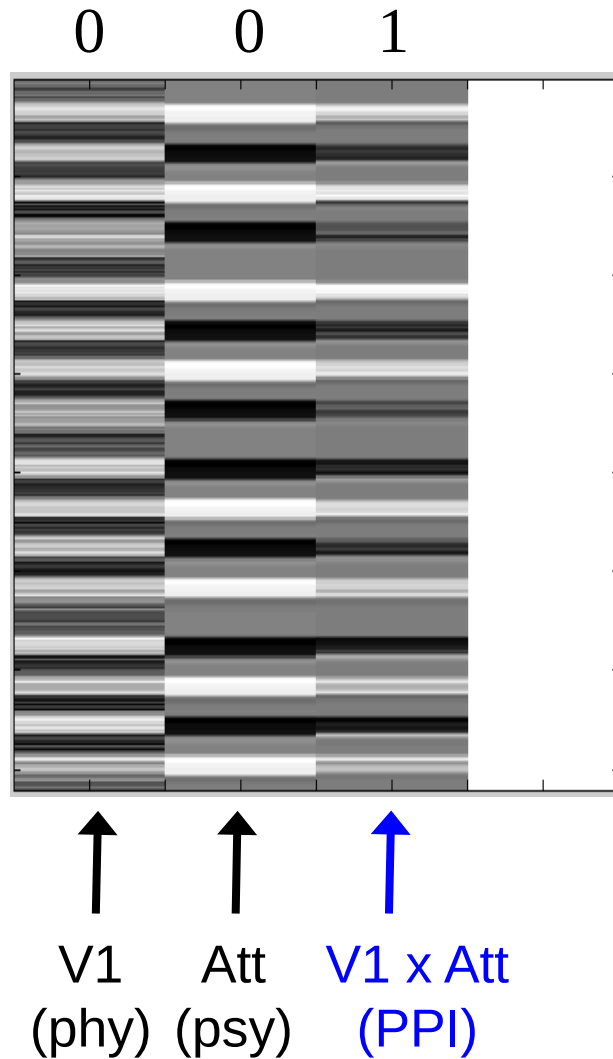
- Bottom line: LSS better when scan noise higher than trial variability (normally the case); LSA better when trial variability higher than scan noise (see “efficiency” notebook)

2. PPIs

Parametric, factorial design, in which one factor is **psychological** (eg attention)
...and other is **physiological** (*viz. activity extracted from a brain region of interest*)



2. PPIs



2. PPI problem with brief psychological factors...



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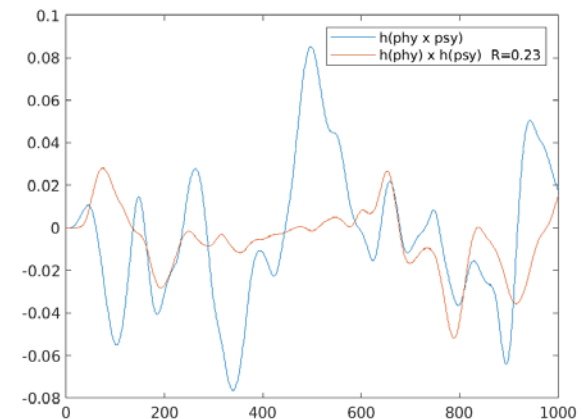
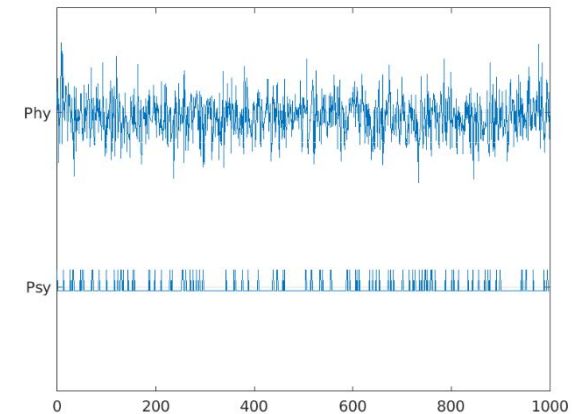
- PPI term is: $(Phy \cdot HRF) \times (Psy \cdot HRF)$
- True neural interaction is: $(Phy \times Psy) \cdot HRF$
- While:

$$(Phy \cdot HRF) \times (Psy \cdot HRF) \approx (Phy \times Psy) \cdot HRF$$

when Psy is low-frequency (e.g, blocked)...

...not case when high-frequency (e.g., event-related)

- So PPI only really suitable for slow (blocked) designs (“gPPI” option, but no true deconvolution – need forward model like in DCM...)
- (Similar problem with “Beta-series regression” – where short SOAs make separation of neural activity tricky...)



Effective-connectivity: Definitions of Causality?

1. Direct experimental interventions (e.g, lesion, drugs)
2. Indirect experimental manipulations (e.g, PPI, DCM)
3. Network model inference (e.g, SEM, DCM)
4. Temporal precedence (e.g, Granger Causality, DCM)
5. ...

Effective-connectivity: Definitions of Causality?



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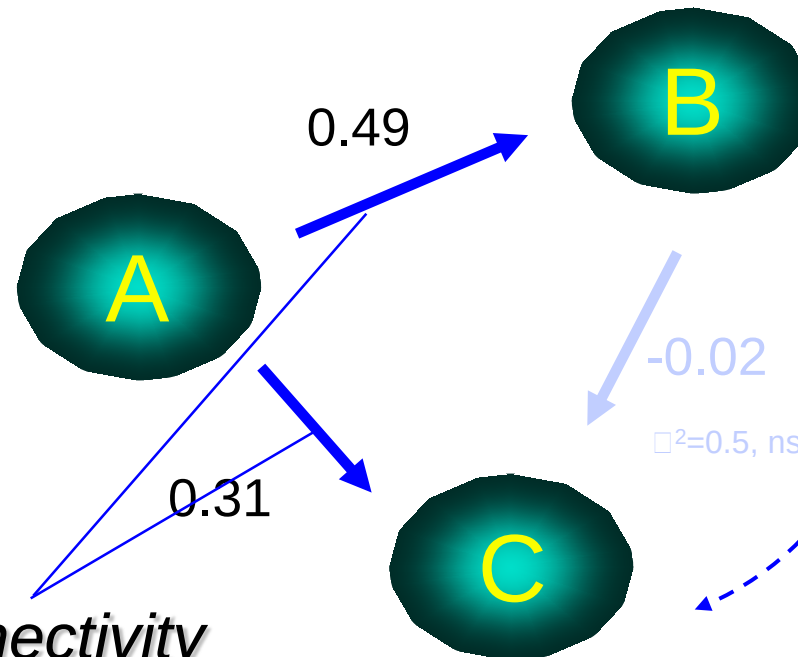
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Effective connectivity

Correlations:

A	B	C
1		
0.49	1	
0.30	0.12	1

*Functional
connectivity*

*Partial Correlation is
approximate way*

3. Explicit Network Models of Causality

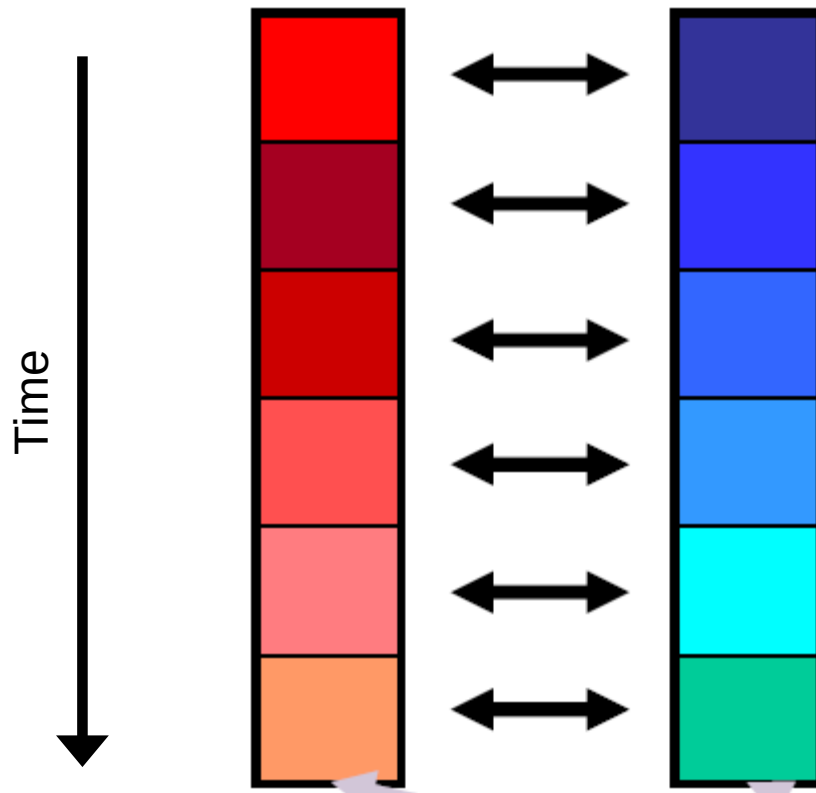
- (Bivariate) correlations do not use an explicit network (graph) model
- Can use *partial* correlation to adjust for all other connections within a graph (Smith et al, 2011; but often requires regularisation and can remove true connections)
- Multivariate methods like Structural Equation Modelling (SEM) can test different network models, by simply comparing *predicted* with *observed* covariance matrices, but...
 - has no dynamical model (stationary covariances)
 - has no neural-BOLD model
 - cannot test some graphs, eg loops (no temporal definition of direction)
 - restricted to classical inference comparing nested models

Effective-connectivity: Definitions of Causality?

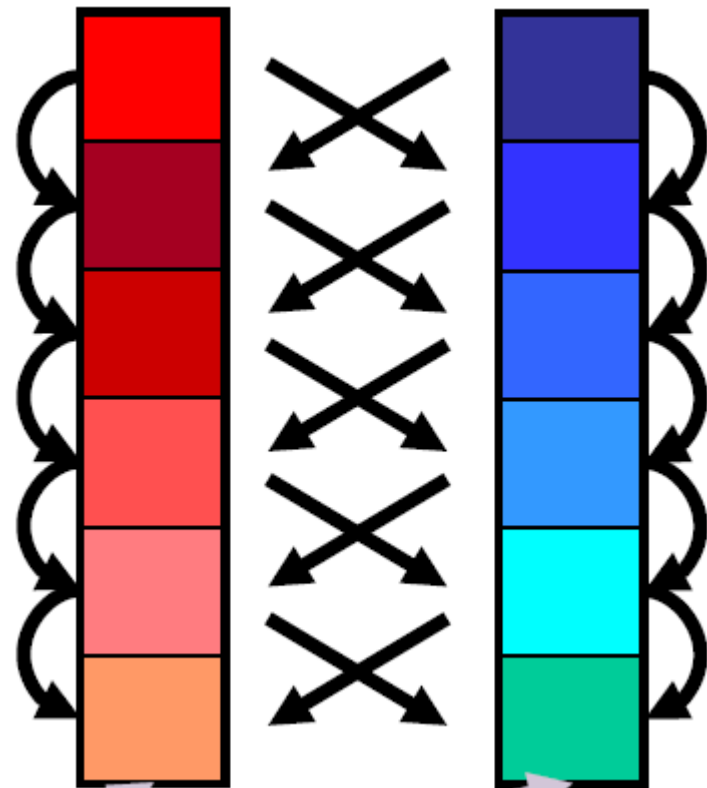
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4. Temporal definition of Causality

Stationary
(correlations, SEM)



Dynamic
(Granger, DCM)



(“unfolding” in time is one way to infer direction of connectivity)

4. Note on temporal causality and fMRI

- Problem with time-based measures of connectivity arises with fMRI: **BOLD** timeseries is not direct reflection of **Neural** timeseries
 - (e.g, peak BOLD response in motor cortex can precede that in visual cortex in a visually-cued motor task, owing to different neural-BOLD mappings)
- This compromises methods like Granger Causality and Multivariate Auto-Regressive models (MAR) that operate directly on fMRI data
(Friston, 2010; Smith et al, 2011)
- (Note that this does not preclude these methods (eg MAR) for MEG/EEG timeseries, assuming these are more direct measures of neural activity)

=> Development of DCM

1. *Dynamic*: based on first-order differential equations

- at level of neural activity, with separate haemodynamic model for fMRI

2. *Causal*: based on explicit directed graph models

3. *Modelling*: designed to test experimental manipulations

- “bilinear” approximation to interactive dynamics

4. (Estimated in a Bayesian context, allowing regularized estimation through priors, formal comparison of any number/type of models, etc...)

Rough comparison of popular methods?



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	Experimental modulation	Temporal/ Dynamical	Network model	Haemodynamic Model (for fMRI)
Correlation / ICA / PCA				
BSR / PPI	Y			
Granger		Y		
SEM			Y	
DCM	Y	Y	Y	Y

Other types of fC (not covered here)



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- **Multi-dimensional** (pattern-based) fC (e.g., Basti et al., 2020, <https://doi.org/10.1016/j.neuroimage.2020.117179>)
- **Dynamic** fC (e.g moving windows or Hidden Markov Models, eg review: <https://pmc.ncbi.nlm.nih.gov/articles/PMC3807588/>)